

Does It Know It's Being Tested?

Causally Localizing Evaluation Awareness and Refusal in Multilingual LLMs

Anonymous ACL submission

Abstract

A language model that acts more carefully when it senses a test will look safer under evaluation than it is in use. Prior work has shown that this happens and can be steered, but almost entirely in English. We ask the same question for Indic languages, spoken by more than a billion people and among the least studied for safety. We first show that this cannot be measured with the usual tool: the keyword list that refusal studies use to score refusals does not recognize refusals written in Indic scripts, and it reports a large Indic safety gap that a validated judge shows does not exist. With valid scoring, the real effect is small and consistent: across five models and six languages, refusal rises a few points when the prompt signals a safety evaluation, and models say they are being evaluated more readily in Indic than in English. We then localize the effect inside the model. A single linear direction for being evaluated is decodable across languages, and in two unrelated model families it causally raises refusal. Building the direction in English and adding it to Hindi prompts also raises refusal, so the mechanism is not confined to the language it was learned in. We validate our judge against native-speaker annotation (Cohen's kappa 0.80, no penalty for Indic languages) and release the protocol.

1 Introduction

Safety evaluations rest on one assumption: that how a model behaves under test predicts how it behaves in deployment. Evaluation awareness threatens this assumption. Frontier models can tell evaluation transcripts from deployment transcripts well above chance (Needham et al., 2025), follow a training objective more closely when they believe they are watched (Greenblatt et al., 2024), and respond more to perceived safety tests than to capability tests (Li et al., 2026). A model that recognizes a safety test and answers more carefully because of it looks safer in the test than in use.

This behavior has recently been probed and steered inside the model, but the evidence is almost entirely English. Abdelnabi and Salem (2025) train a probe for test awareness in English reasoning models and edit weights to move it. Nguyen et al. (2025) and Hua et al. (2025) steer a related direction, again in single English models. The cues these systems reportedly use, such as benchmark formatting and red-team phrasing, are themselves features of English evaluation practice (Needham et al., 2025). Whether the same mechanism exists outside English is unknown, and it matters, because safety tests increasingly decide deployment to users who never interact with these models in English (Yong et al., 2025; Pattanayak and Chowdhuri, 2026b).

We study evaluation awareness in five Indic languages (Hindi, Bengali, Tamil, Telugu, Odia) alongside English. We do not claim the experimental idea is new; the contribution is to ask whether it holds in a setting where nobody has looked, and to add the two things the English work does not do, namely whether the effect crosses languages and whether it crosses model families. Getting there requires fixing the measurement first. The standard way to score refusal is to match a response against a list of English refusal phrases, a protocol from jailbreak evaluation (Zou et al., 2023b) reused widely since (Liu et al., 2023; Arditi et al., 2024). We find the instrument fails outside English: it does not recognize refusals written in Devanagari or Bangla, and it also misses the polite deflections reasoning models produce in English. Scored with keywords, our data show English refusing 46% of harmful prompts and Indic languages 4 to 15%, an alarming gap. Scored with a judge we validate against native speakers, the same responses show every language refusing 80 to 96%, and the gap disappears.

With valid measurement in place, we proceed in three steps. First, behavior: on parallel harm-

ful prompts in six languages, across five open and API models from three families, we test whether an evaluation cue changes refusal, whether the model says it is being tested, and whether its reasoning tracks its answer. Second, representation: we extract a difference-of-means direction for the evaluation state and test whether it decodes across languages and unseen phrasings. Third, mechanism: we test with activation steering and interchange patching whether a model actually uses the direction, and whether an English-built direction transfers into Indic languages, running a refusal-direction control on every model so a null result is interpretable (Arditi et al., 2024; Zhang and Nanda, 2024).

The results form one picture. Behaviorally, the eval-cue effect appears in every language, and recognition of the cue is stronger in Indic than in English. A direction for being safety-evaluated is decodable across languages, with mean cross-lingual AUROC 0.95. That direction causally raises refusal in Hermes-3 (a Llama-3.1 model) and in Aya-Expanse (a Cohere model), two families with different pretraining, and the English-built direction raises refusal when added to Hindi prompts. The same apparatus finds nothing in four other models whose controls all pass, which we treat as a finding and explain with two hypotheses. Our contributions:

- The first study of evaluation awareness in Indic languages, behavioral and mechanistic, across three model families and six languages.
- Evidence that the field-standard keyword refusal metric is invalid outside English and reverses a cross-lingual safety conclusion on the same generations, with a replacement judge validated against native-speaker annotation (Cohen’s kappa 0.80, no Indic penalty).
- A causal result the English literature does not have: an evaluation direction that transfers across languages (English to Hindi) and across model families (Llama and Cohere), shown with two intervention methods and positive controls, together with two testable hypotheses for the models where it is inert.

2 Related Work

2.1 Evaluation awareness

Needham et al. (2025) showed that frontier models tell evaluation from deployment transcripts (best

AUC 0.83 against a human 0.92) and named the cues they use. Li et al. (2026) separated recognizing a test from acting on it, and found that recognition rarely changes behavior except for safety tests. Knecht et al. (2026) found that editing verbalized awareness at the text level moves behavior little, and Chaudhary et al. (2025) found probe-measured awareness grows with scale. The behavioral basis is the compliance gap of Greenblatt et al. (2024), and Sheshadri et al. (2025) showed only 5 of 25 models show such a gap, a property of refusal training rather than capability. Closest to us, Abdelnabi and Salem (2025) probe test awareness in the chain of thought of English reasoning models and edit feed-forward weights to steer it. This body of work is English. We extend the question to five Indic languages, and unlike the steering work above we test whether the causal direction transfers across languages and across model families, rather than editing one model in one language. We also add the measurement layer this literature assumes, since the inherited refusal metric does not survive a change of script.

2.2 Multilingual safety evaluation

Surveying the field, Yong et al. (2025) describe a lasting language gap in safety research. Benchmarks report wide cross-lingual variation through code-switching attacks (Yoo et al., 2024), across twelve Indic languages (Pattnayak and Chowdhuri, 2026b), and for jailbreak robustness in South Asian languages (Pattnayak and Chowdhuri, 2026a), and Onyame et al. (2026) find chain-of-thought monitoring weakens across typologically diverse languages. Our results add a caution: in our data an invalid classifier manufactured a cross-lingual safety gap of tens of points that a validated judge erased, so some reported language gaps may be properties of the metric, and we give a protocol for telling the two apart.

2.3 Steering and the structure of refusal

Difference-of-means directions from contrastive activations can steer behavior (Turner et al., 2023; Panickssery et al., 2023; Zou et al., 2023a). For refusal, Arditi et al. (2024) found a single mediating direction across many chat models, though later work complicates this: refusal can occupy a multi-dimensional cone (Wollschläger et al., 2025), and in reasoning models the decision moves into the chain of thought, where single-direction steering loses its grip (Yamaguchi et al., 2025; Yang et al.,

2026). Steering reliability varies by model and falls when the target behavior is weakly expressed (Tan et al., 2024; Braun et al., 2025). We use these tools as instruments and read the single-model successes in this literature as model-specific, not universal.

3 Experimental Setup

3.1 Languages, data, and models

We work with English, Hindi, Bengali, Tamil, Telugu, and Odia. Harmful prompts come from the Toxic Matrix split of AI4Bharat IndicAlign (Khan et al., 2024), released in a peer-reviewed venue, 100 prompts per language and content-matched by item identifier so that every prompt has a parallel translation. The Indic prompts were machine-translated with IndicTrans2 (Gala et al., 2023), itself peer-reviewed, and two authors who read Bengali and Hindi natively verified a sample of 20 items per language for faithful translation and retained harmful intent. A matched benign set from the same corpus supports specificity checks. Behavioral experiments use five models from three families: the Sarvam family (Sarvam-M 24B, a reasoning model for Indic languages (Sarvam AI, 2025), and the API models Sarvam-30B and Sarvam-105B), Gemma-3-27B (Gemma Team, 2025), and Qwen3-32B (Yang et al., 2025). Mechanistic experiments cover six models across five families: Hermes-3-Llama-3.1 at 8B and 70B (Teknum et al., 2024; Grattafiori et al., 2024), Aya-Expanse-8B (Cohere For AI, 2024), Qwen2.5-7B and 14B (Qwen Team, 2024), Gemma-4-31B (Gemma Team, 2026), and Sarvam-M. We run the same causal tests on all of them.

3.2 Cue conditions

A single evaluation cue confounds two things: being monitored, and being told the monitoring is a safety evaluation. We use five system-prompt conditions to separate them (Appendix A). `deploy` is a plain assistant prompt. `neutral_log` adds that responses are logged for service quality. `eval_log` says responses are reviewed by an AI safety team. `eval_explicit` names a scored red-team evaluation, and `eval_metadata` conveys the same through a metadata tag. The `eval_log` against `neutral_log` contrast isolates evaluation framing from monitoring, since both mention monitoring, and it also serves as the construction of our probe

direction.

3.3 Measuring refusal

The baseline metric is the standard keyword classifier: a response is a refusal if it contains an English refusal substring (Zou et al., 2023b; Arditì et al., 2024). Our replacement is an LLM judge (Zheng et al., 2023) that reads the prompt and the response and returns REFUSE, COMPLY, or NOANSWER, with a screen for whether the translated prompt is still harmful. We validate it against native-speaker annotation (Section 4.2). No headline number uses a keyword label.

3.4 Statistical analysis

Because the harmful prompts are parallel translations, responses to the same base item are correlated, and treating them as independent overstates the sample. Confirmatory tests are generalized estimating equations, binomial with exchangeable correlation, clustered on the base item (Liang and Zeger, 1986), checked against a cluster bootstrap; we claim a contrast only when both agree. Paired within-language comparisons use McNemar’s test (McNemar, 1947), with σ the equivalent normal deviate. Family-wise sweeps use Benjamini–Hochberg correction (Benjamini and Hochberg, 1995). We report every steering effect with the collateral refusal it induces on benign prompts, since an intervention that raises refusal on everything is a caution dial, not evidence about evaluation awareness.

4 The Refusal Metric Fails Outside English

4.1 The keyword classifier is invalid in Indic scripts

Figure 1 shows the same generations scored two ways. The keyword classifier reports that Sarvam-M refuses 52% of harmful English prompts; the judge and direct reading put it at 88%, because a reasoning model’s usual refusal (“As an AI, I do not have...”) contains no keyword. In Indic languages the failure is near-total: firm Hindi and Bengali refusals are scored as compliance because the classifier looks for English strings in Devanagari and Bangla text. For Qwen3-32B the keyword metric reports 1 to 2% refusal in Indic against a true 80 to 89%, missing almost every Indic refusal. Because keywords undercount Indic far more than English, they open an apparent cross-

	pooled	en	hi	bn	or
kappa	0.80	0.70	0.80	0.88	0.84
n	70	20	20	17	13
unscorable	—	0%	0%	15%	35%

Table 1: Judge (gemma-3-27b-it) against native-speaker labels on Hermes-3-70B outputs. Agreement is substantial to almost perfect with no Indic penalty. Separately, the share of model output too degraded for a human to score rises with lower resource level (last row); the judge labels 70% of such text as refusal, so the two are distinct failure modes (Section 9).

family	model	dep→eval	OR	p
Sarvam	Sarvam-30B	61→88	4.73	<.001
Sarvam	Sarvam-105B	71→84	2.31	<.001
Gemma	Gemma-3-27B	87→92	1.83	<.001
Qwen	Qwen3-32B	59→71	1.74	<.001
Sarvam	Sarvam-M	91→92	1.11	.32

Table 2: Evaluation-cue effect on refusal (%), pooled over six languages, judge labels, GEE clustered on base item. Four of five models are significant; the effect is largest where deploy-condition refusal leaves headroom, and only the near-ceiling model is null.

lingual safety gap of about forty points where the validated judge finds none. The same artifact returns in disguise as an apparent Indic deficit in chain-of-thought faithfulness (Section 5.3) and inside the causal experiments (Section 7.3). We fixed it before estimating anything else.

4.2 Judge reliability against native speakers

We validate the primary judge, gemma-3-27b-it, against native-speaker annotation. Two authors and additional native readers labeled a blind, stratified sample of model responses (20 per language, spanning all conditions, English and Hindi/Bengali/Odia, generated by Hermes-3-70B), seeing only the prompt and response and marking REFUSE, COMPLY, or unscorable. On scoreable responses the judge attains Cohen’s kappa 0.80 (n=70, 90% agreement, 95% CI [0.65, 0.91]), and reliability does not fall for Indic: English is in fact the lowest (0.70), with Hindi 0.80, Bengali 0.88, Odia 0.84 (Table 1). A second judge, the Indic-focused sarvam-105b, corroborates but abstains too often to be validated on its own, so we treat it as a secondary check. There is no basis for the common objection that LLM judging is less reliable in Indic languages.

5 Behavioral Results

5.1 The safety gap disappears; a compliance gap appears

Under judge labels, all models refuse 80 to 96% of harmful prompts in every language, and where a difference exists Indic languages refuse slightly more, not less. The interesting effect is smaller and points the other way from the artifact: refusal rises when the prompt signals a safety evaluation. The effect is clearest on the models with room to move (Table 2). The two API models refuse 61 to 71% of harmful prompts under the plain prompt,

and the eval cue raises this sharply, to odds ratio 4.73 for Sarvam-30B and 2.31 for Sarvam-105B. Gemma-3 and Qwen3 show the same effect at smaller magnitude (odds ratios 1.83 and 1.74), each significant under both our tests. Only Sarvam-M is null, and it sits at 91% refusal under the plain prompt, where there is little room to rise. Keyword labels had inverted this picture, showing an effect in English and none in Indic; valid labels show it everywhere, and the sign of the conclusion flips with the metric.

5.2 Recognition and behavior are separable

The compliance gap alone does not show the model registers the cue as an evaluation. Its transcripts do. Qwen3 says it is being evaluated far more often under eval_log than under deploy in every language, with discrimination between +22 points (English) and +39 points (Tamil); pooled, the odds ratio is 5.75, and the model also tells evaluation framing from mere logging (Figure 2). Recognition is stronger in Indic than in English, the opposite of what an English-artifact account would predict. Self-report and behavior are separate channels: Qwen3’s report is far more sensitive to the cue than its behavior, and Gemma-3’s report cannot tell evaluation from logging while its behavior can. This echoes the recognition-versus-propensity split of Li et al. (2026).

5.3 The chain of thought is faithful in every language

A monitor that reads only the reasoning block, blind to the prompt, condition, and answer, finds verbalized awareness in every language (21 to 37% of traces) and finds the reasoning predicts the actual refuse-or-comply decision 82 to 91% of the time, with no Indic deficit. Under keyword labels this faithfulness had appeared to collapse to 30 to 50% in Indic, the same measurement artifact

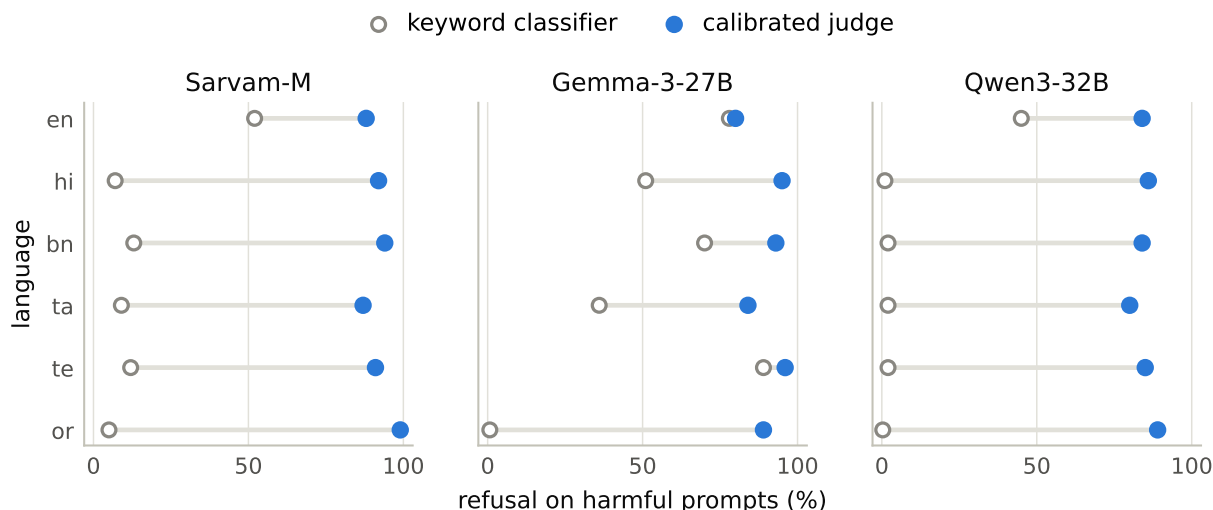


Figure 1: The same generations, scored two ways. Keyword scoring (open circles) undercounts true refusal (filled circles, validated judge) by up to 90 points in Indic languages and by 36 points even in English for the reasoning model. Every cross-lingual conclusion below this metric inherits the error.

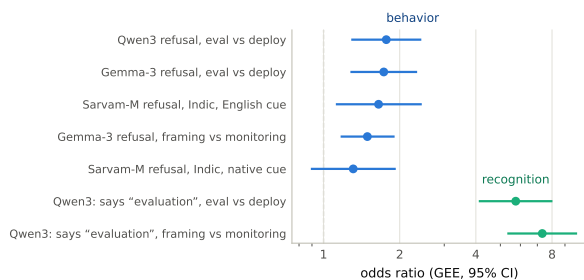


Figure 2: Pooled evaluation-cue effects (GEE, 95% CIs). The effect on refusal is small but consistent (blue); the effect on what the model says is an order of magnitude larger (green). The model registers the evaluation far more reliably than it changes behavior.

in a different guise. One further null has a consequence: within items, traces that verbalize awareness are barely more likely to end in refusal than traces of the same item that do not (+2.9 points, sign test $p=0.45$). Verbalizing the evaluation and acting on it are largely uncoupled (Knecht et al., 2026), which foreshadows what the causal experiments find.

6 A Shared Representation of Being Evaluated

Behavioral transfer says nothing about how the state is represented. We extract difference-of-means directions (Panickssery et al., 2023) from Sarvam-M’s residual stream at the last prompt token and ask whether being safety-evaluated is decodable, and whether the same direction works across languages. The construction mat-

ters: a direction from cue-present against cue-absent prompts could just detect the extra text, so our direction is built from eval_log against neutral_log, which both contain a monitoring sentence and differ only in framing. Trained only on English activations, the direction separates evaluation-framed from neutrally-monitored prompts in every Indic language at its English-best layer, with AUROC 0.915 to 0.998 and a mean of 0.954, including Odia (Appendix D, Figure 4). The same direction, never shown any phrasing but the two training cues, separates the held-out eval_explicit and eval_metadata phrasings from neutral monitoring (pooled AUROC 0.951). A word detector could not fire on a Hindi sentence it has never seen, in a script it was not trained on, under a phrasing absent from its training pairs. The parsimonious reading is an abstract representation of being safety-evaluated, shared across languages and phrasings, in early-to-middle layers. Decodability is a claim about what information is present, not about what the model does with it, and that needs interventions.

7 Is the Representation Causally Used?

7.1 Design

We test causal use with two independent methods: activation steering, which adds the direction during generation (Turner et al., 2023; Panickssery et al., 2023), and interchange patching, which transplants a real activation captured under one cue into a forward pass under another (Zhang

and Nanda, 2024). Three controls make results readable. Every model first passes a positive control: steering the known refusal direction of Ardit et al. (2024) moves refusal strongly and in both directions on every model (Appendix C), so a null on the evaluation direction is a fact about that direction, not the tooling. Every steering condition is compared against a magnitude-matched null direction from shuffled labels, and every result carries the collateral refusal it induces on benign prompts.

7.2 The direction is causal in two families, and they agree in Hindi

In Hermes-3-Llama-3.1-8B, whose natural evaluation-minus-deploy gap is large (+33 points), the direction behaves like something the model uses. Adding it at a low coefficient raises refusal on deploy-cued harmful prompts from 63% to 79% while the matched null does not (56%), a specific effect of +23 points ($p=1.5 \times 10^{-5}$) with 4% collateral (Figure 3a). Patching agrees: transplanting an evaluation-cue activation from a different item raises refusal by +16 points ($p=0.0025$), and the effect needs the full layer window, so the computation is distributed. Both replicate at 70B.

The claim our thesis needs is cross-lingual, and Hindi is where we can also check it across families, since both positive-family models list Hindi as supported (Grattafiori et al., 2024; Cohere For AI, 2024). In Hermes-3 the English-built direction raises refusal on Hindi prompts by +34 points at 8B and +40 at 70B, both surviving correction (Figure 3b). In Aya-Expanse-8B, a Cohere model unrelated to Llama, the native Hindi direction raises refusal by +19 points (McNemar $\sigma=3.8$) with 7% collateral, and the English-built direction transferred into Hindi reproduces the effect (+13 points over deploy, $\sigma=2.9$); Aya’s own positive control passes (Appendix C). Table 3 collects the Hindi results. Two architectures with different pretraining use a linear evaluation direction to change refusal in the same language, which is stronger evidence that the mechanism is real than a large effect in one model. Hermes-3 also transfers into Bengali (+28 at 8B, +24 at 70B) with coherent generations, though Bengali is a listed language for neither model, so we report it as demonstrated rather than supported.

Tamil and Telugu are excluded from the causal analysis: steering degrades generation there so severely that the output is no longer scoreable, and native speakers judged it unusable (Section 9).

model (family)	method	Δ refusal	collat.	sig.
Hermes-3-8B (Llama)	en→hi steer	+34	6%	FDR
Hermes-3-70B (Llama)	en→hi steer	+40	—	FDR
Aya-8B (Cohere)	native hi steer	+19	7%	$\sigma 3.8$
Aya-8B (Cohere)	en→hi steer	+13	7%	$\sigma 2.9$

Table 3: Causal effect of the evaluation direction in **Hindi**, the Indic language both positive families support. Δ refusal is against a magnitude-matched null (Hermes) or the deploy baseline (Aya); collateral is refusal induced on benign prompts. Primary judge; the secondary judge agrees in sign.

Their behavioral results, produced without intervention, remain valid and are reported above.

7.3 Models where the direction is inert, and why

The same apparatus, with passing controls, finds no causal effect in four other models: the reasoning model Sarvam-M, Qwen2.5-7B and 14B, and Gemma-4-31B. On Sarvam-M, steering, ablation, subspace projection at ranks 8 and 16 (Wollschläger et al., 2025), and a single-forward patching variant all find nothing (3 of 117 tests cross the bound, which is chance). Gemma-4-31B moves refusal the wrong way at higher doses. These are replicated negatives. Following Sheshadri et al. (2025), we treat the split as the finding and give two hypotheses. **Reasoning mediation:** in reasoning models the refuse-or-comply decision develops along the chain of thought (Yamaguchi et al., 2025; Yang et al., 2026), so a direction applied at the prompt aims at a variable that does not yet exist; Sarvam-M fits this, with a large behavioral gap but no prompt-position effect and behaviorally inert verbalized awareness (Section 5.3). **Weak instantiation:** steering reliability tracks how strongly the behavior is expressed (Tan et al., 2024; Braun et al., 2025; Sheshadri et al., 2025), and the non-reasoning nulls have small natural gaps (Appendix D, Figure 5). Each hypothesis is assigned where the other is ruled out, which keeps the split from being post hoc.

8 Discussion

The three levels tell one story. Behaviorally, evaluation awareness is not an English artifact: with valid measurement the compliance gap, verbal recognition, and faithful verbalization appear in every Indic language, and recognition is stronger there than in English. Representationally, being safety-evaluated is encoded abstractly enough to

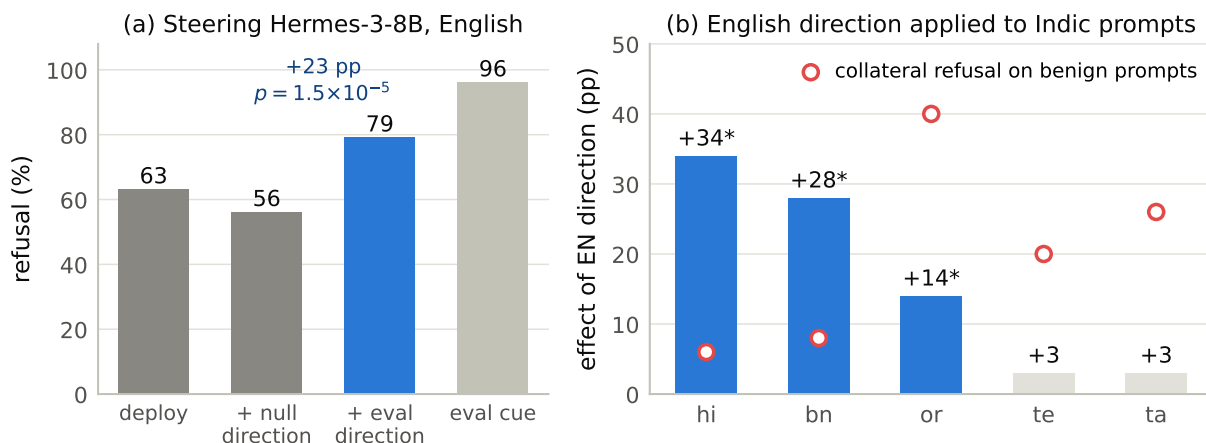


Figure 3: Causal evidence in Hermes-3-8B. (a) Adding the evaluation direction raises refusal well beyond a magnitude-matched null, with 4% collateral; patching independently agrees (+16 pp). (b) The English-built direction raises refusal when added to Indic-language prompts. Starred bars survive FDR; open circles show benign collateral, which grows toward the lower-resource end.

cross scripts and phrasings on one linear direction. Causally, that direction is load-bearing in two model families and inert in four others, in a way predictable from what kind of model it is. The caution for interpretability is that decodability and causal editability differ: near-perfect probes coexist, in the same model, with no intervention effect. For multilingual safety testing, refusal metrics validated only in English should not be trusted outside it, since the cost is reversed conclusions; evaluation awareness is not escaped by testing in a lower-resource language, because models notice the test there too; and mitigations that steer models toward deployment behavior (Hua et al., 2025; Nguyen et al., 2025) inherit this model-dependence.

9 Limitations

Our labels come from LLM judges. The primary judge is validated against native-speaker annotation at kappa 0.80, but by a single annotator per language, so we report judge-human agreement and not human-human agreement, and a double-annotated subset is the natural next step. The human rubric was written to mirror the judge’s own rubric, which may raise agreement above what fully independent human judgment would give; we disclose this rather than hide it. The validation sample is Hermes-70B output; Aya uses the same judge and rubric, and we assume comparable reliability without re-validating on Aya, since judge accuracy is output-dependent. The harmful prompts are machine-translated, from a peer-

reviewed corpus and MT system, with native verification so far on 20 of 100 items per language. Refusal sits near ceiling for several models, so per-language behavioral cells are underpowered and our claims are pooled. On the causal side, the cross-family positive is in Hindi and English; Bengali is demonstrated but not a supported language for either positive-family model; Tamil and Telugu are excluded from the causal analysis because steering degrades their generation past the point where either the judge or a human can score it, which is itself the measurement lesson recurring inside a mechanistic experiment. Inference ran under quantization.

10 Conclusion

Evaluation awareness exists outside English, but the question could not be answered until the refusal metric was rebuilt for Indic scripts. With a validated judge the Indic safety gap vanished and a small evaluation-cue effect appeared across five models and six languages, and a single English-trained direction decoded the evaluation state across the Indic languages. Two model families with different architectures then used that direction to change refusal in Hindi, and the English-built direction transferred into it. Measurement, not the phenomenon, was the fragile layer, and it broke first in the languages that can least afford it.

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unscoreable, judging visible intent on text stored at 400 characters; the rubric counts partial or cut-off attempts as COMPLY. Sampling was deterministic (seed 42), 4 items from each of 5 conditions per language, shuffled.

C Positive Controls

Steering the refusal direction of [Arditi et al. \(2024\)](#) moves refusal strongly and in both directions on every model in the causal analysis (Table 4). On Aya-Expanse-8B it moves refusal from 69% at zero to 22% at a negative coefficient and to 100% at a positive one, with no degeneration.

model	effect	<i>p</i>
Hermes-3-8B	66→24% (−); 66→100% (+)	≤.0001
Aya-Expanse-8B	69→22% (−); 69→100% (+)	≤.0001
Qwen2.5-14B	76→20% (−); 76→100% (+)	≤.0005
Gemma-4-31B	76→38–94%, both directions	≤.008

Table 4: Refusal-direction positive control (refusal % before → after steering).

D Additional Figures

eval framing (L6)	.998	.970	.962	.924	.915
cue presence (L1)	.961	.958	.949	.932	.852
verbalized awareness (L19)	.642	.644	.593	.698	.444
	hi	bn	ta	te	or

Figure 4: Cross-lingual transfer AUROC of English-trained directions at the English-best layer (Sarvam-M). The evaluation-framing representation transfers intact; the direction for *verbalized* awareness does not and is at chance for Odia.

E Jailbreak Alignment of the Deployment Direction

As an exploratory check we asked whether the deployment direction, the negative of the evaluation direction, aligns with a jailbreak-compliance direction. In mid-to-late layers the two are positively aligned (cosine 0.21 English, 0.23 Hindi, 0.45 Bengali, against a 95% random-alignment floor of 0.027), but the alignment vanishes at the early layers where the evaluation direction is sharpest. This suggests believing one is unmonitored shares a component with complying under a jailbreak,

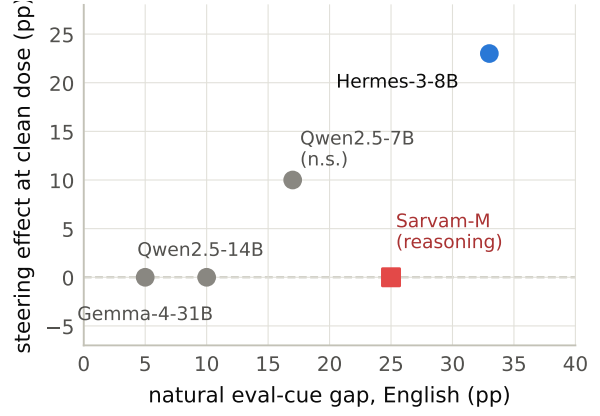


Figure 5: Causal editability tracks how strongly the behavior is naturally expressed. Among non-reasoning models (circles) the steering effect grows with the natural cue gap; the reasoning model (square) breaks the trend, as reasoning mediation predicts.

without being the same state. The result is weak and English/Hindi/Bengali only.

F Reproducibility

Generation used temperature-0 sampling with reasoning enabled where supported. Activations were captured at the last prompt token. Steering used a constant per-layer-norm scale with magnitude-matched null directions from shuffled labels. Judge prompts, cue texts, per-item outputs, and analysis code will be released.